CY1001
Summer 2026
Soumen Ghosh

1. Calculate the probability that a particle in a one-dimensional box of length a is found to be between $a/2$ and $a/4$.
2. Show that any linear combination of the complex functions e^{iax} and e^{-iax} is an eigenfunction of the kinetic energy operator.
3. Show that the $n = 1$ and $n = 2$ wave functions of particle-in-a one-dimensional box are orthogonal to each other.
4. Normalized wave function for $l = 0$ and $m = 0$ state of a rigid rotor, $Y_{m=0}^{l=0}(\theta, \phi) = A$ where A is the normalization constant. Determine A.
5. Calculate average energy ($\langle \hat{H} \rangle$) and standard deviation of the energy $\left(\sigma_E = \sqrt{\langle \hat{H}^2 \rangle - \langle H \rangle^2}\right)$ for a normalized superposition state $\Psi = C_1\phi_1 + C_2\phi_2$ where $\hat{H}$ is the Hamiltonian, ϕ_1 and ϕ_2 are orthonormal eigenfunction of $\hat{H}$ with eigenvalues E_1 and E_2, C_1 and C_2 are the complex coefficients.
6. Calculate the probability of finding the electron within one Bohr radius (a_0) in the hydrogen atom described by the *1s*-wavefunction,

$$\Psi_{1s} = \frac{1}{\sqrt{\pi}} \left(\frac{1}{a_0}\right)^{3/2} exp\left(\frac{-r}{a_0}\right)$$